

# HW #3

1. The a)  $H_2O$   $p = 100 \text{ psi}$   $T = 327.86^\circ F$

$u = 500 \text{ Btu/lb}$   $u_g = 1105.8$   $u_f = 298.3$   $v_g = 4.434$   $v_f = 0.0174$   $h_g = 1187.8$   $h_f = 298.6$

$x = 0.25$

$$x = \frac{500 - 298.3}{1105.8 - 298.3} = 0.25$$

$$0.25 = \frac{v - 0.0174}{4.434 - 0.0174} \Rightarrow v = 1.1 \text{ ft}^3/\text{lb}$$

$$0.25 = \frac{h - 298.6}{1187.8 - 298.6} \Rightarrow h = (0.25(1187.8 - 298.6)) + 298.6 = 520.4 \text{ Btu/lb}$$

b)  $T = 40^\circ F$

used Matlab for the rest of these, everything came from quality formula and tables

$$p = 73.3 \text{ lbf/in}^2$$

$$u = 327.65 \text{ Btu/lb}$$

$$v = 1.9976 \text{ ft}^3/\text{lb}$$

c)

\*Thank you letting me see some time\*

$$T = 480^\circ C$$

$$u = 3030 \text{ kJ/kg}$$

$$h = 3350 \text{ kJ/kg}$$

d)  $100^\circ F$

$$p = 1.3529 \text{ lbf/in}^2$$

$$v = 0.145 \text{ ft}^3/\text{lb}$$

$$h = 216 \text{ kJ/kg}$$

2. Sat <sup>H<sub>2</sub>O</sup> water

$$\begin{array}{l} \text{1} \\ T = 180^\circ\text{C} \\ v = 0.1941 \text{ m}^3/\text{kg} \\ R = 10.02 \text{ bar} = 1.002 \cdot 10^4 \text{ Pa} \\ \boxed{1.002 \text{ MPa}} \end{array} \quad \text{2} \quad \begin{array}{l} v = 0.1941 \text{ m}^3/\text{kg} \\ T = 100^\circ\text{C} \end{array}$$

$$x = \frac{v - v_f}{v_g - v_f}$$

$$x = \frac{0.1941 - 1.0435 \cdot 10^{-3}}{1.673 - 1.0435 \cdot 10^{-3}} = \boxed{0.1155}$$

3.  $m = 2 \text{ kg}$   
 $p = 300 \text{ kPa} \rightarrow \text{constant}$   
 (3 bar)

Initially sat. H<sub>2</sub>O water

$$\begin{array}{l} \text{State 1 (Sat)} \\ m = 2 \text{ kg} \\ p = 300 \text{ kPa} = 3 \text{ bar} \\ \boxed{T = 133.6^\circ\text{C}} \\ v = 0.6058 \text{ m}^3/\text{kg} \\ V = (0.6058)(2) \\ = 1.2116 \text{ m}^3 \end{array} \quad \frac{p_1 v_1}{T_1} = \frac{p_2 v_2}{T_2} \quad T =$$

$$\begin{array}{l} \text{State 2} \\ m = 2 \text{ kg} \\ p = 300 \text{ kPa} = 3 \text{ bar} \\ v = 2.064 \text{ m}^3/\text{kg} \\ V = 2.064 \text{ m}^3 \\ v = \frac{2.064 \text{ m}^3}{2 \text{ kg}} = 1.032 \text{ m}^3/\text{kg} \end{array}$$

$$\frac{3(0.6058)}{133.6} = \frac{3(1.032)}{T_2} \Rightarrow \boxed{T_2 = 227.59^\circ\text{C}}$$

80%

$$w = -p \Delta V = m p (v_2 - v_1) = 300(2.064 - 1.2116) = \boxed{-255.72 \text{ kJ}}$$

4

State 1

$$T_1 = 60^\circ\text{C} = 333.15\text{ K}$$

$$P_1 = 1\text{ bar} = 1 \cdot 10^5\text{ Pa}$$

$$R = 8.314\text{ kJ/mol}\cdot\text{K}$$

$$m = 1\text{ kg}$$

State 2

$$T_2 = 30^\circ\text{F} = 303.15\text{ K}$$

$$P_2 = 2\text{ bar} = 2 \cdot 10^5\text{ Pa}$$

$$R = 8.314\text{ kJ/mol}\cdot\text{K}$$

$$m = 1\text{ kg}$$

$$W = -dC\text{ kJ}$$

80%

$$\text{State 1: } P_1 V_1 = m R T_1$$

$$(1 \cdot 10^5) V_1 = (1)(8.314)(333.15)$$

$$V_1 = 2.77 \cdot 10^{-3}\text{ m}^3/\text{kg}$$

$$X_1 = \frac{2.77 \cdot 10^{-3} - 0.9488 \cdot 10^{-3}}{0.0114 - 0.9488 \cdot 10^{-3}} = 0.174$$

$$0.174 = \frac{u_1 - 135.82}{256.81 - 135.82} \Rightarrow u_1 = 156.87\text{ kJ/kg}$$

AND

$$U_1 = 156.87\text{ kJ}$$

$$\text{State 2: } P_2 V_2 = m R T_2$$

$$(2 \cdot 10^5) V_2 = (1)(8.314)(303.15)$$

$$V_2 = 0.0126\text{ m}^3/\text{kg}$$

$$X_2 = \frac{0.0126 - 0.8417 \cdot 10^{-3}}{0.0265 - 0.8417 \cdot 10^{-3}} = 0.45$$

$$0.45 = \frac{u_2 - 90.84}{243.1 - 90.84} \Rightarrow u_2 = 159.357\text{ kJ/kg}$$

AND

$$U_2 = 159.357\text{ kJ}$$

$$Q + W = \Delta U + \Delta KE + \Delta PE$$

$$\Delta U = U_2 - U_1 = 159.357 - 156.87 = 2.487\text{ kJ}$$

$$Q - 20 = 2.487 \rightarrow$$

$$Q = 22.487\text{ kJ}$$



5. Closed rigid tank  $\rightarrow v_1 = v_2$

$$m = 1.5 \text{ kg}$$

State 1: MWV

$$T_1 = 20^\circ\text{C}$$

$$v = 0.8919 \text{ m}^3/\text{kg}$$

$$p_1 = 31.19 \text{ kPa} \quad (\text{from table A-2})$$

$$u_1 =$$

State 2: SWV

$$T_2 = 120^\circ\text{C}$$

$$v = v_g = 0.8919 \text{ m}^3/\text{kg}$$

$$u_2 = u_g = 2528.9 \text{ kJ/kg} \quad (\text{from table A-2})$$

$$m x = \frac{0.8919 - 1.0228 \cdot 10^{-3}}{5.042 - 1.0228 \cdot 10^{-3}} = 0.1767$$

$$0.1767 = \frac{u_1 - 292.95}{2469.6 - 292.95} \rightarrow u_1 = 677.56 \text{ kJ/kg}$$

$\nearrow$  no DV, negligible

Final heat transfer:  $Q + W = \Delta U + \Delta KE + \Delta PE$

$$Q = \Delta U$$

$$Q = 2777.01 \text{ kJ}$$

$$u_1 = 677.56 \cdot 1.5$$

$$u_2 = 2528.9 \cdot 1.5$$

$$\Delta U = u_2 - u_1$$

$$\Delta U = 1.5 (2528.9 - 677.56)$$

$$= 2777.01 \text{ kJ}$$

6. Refrigerant 22  $\rightarrow$  Table A-7

$$m = 2 \text{ kg}$$

$$x = 0.6$$

$$PV = mRT$$

$$p = 10 \text{ bar} = 1 \cdot 10^6 \text{ Pa} = 1 \cdot 10^6 \text{ Pa}$$

$$T \text{ (from table)} = 23.40^\circ\text{C} = 273.15 + 23.40 = 296.55 \text{ K}$$

Need kelvin and Pa  $\rightarrow$  standard units  
for  $R = 8.314$

Saturated Vapor

$$(1 \cdot 10^6) v = (2) (8.314) (296.55)$$

$$v = 0.0616 \text{ m}^3/\text{kg}$$

$$0.6 = \frac{v - 0.8352 \cdot 10^{-3}}{0.0236 - 0.8352 \cdot 10^{-3}}$$

$$v = 0.0145 \text{ m}^3/\text{kg}$$

$$\frac{v_g}{v} = \frac{0.0236}{0.0616}$$

$$= 0.383$$

$$\text{or } 38.3\%$$

$$\frac{v}{v} = \frac{0.0145}{0.0616} = 0.235 \text{ or } 23.5\%$$

Specific Volume

7.  $C.2g He = m = 2 \cdot 10^{-4} kg He$

State 1:  
 $T_1 = 27^\circ C = 300.15 K$   
 $P_1 = 1.3 bar = 1.3 \cdot 10^5 Pa$

$(1.3 \cdot 10^5)$

~~$z_c = \frac{P_c v_c}{R T_c}$~~   
 ~~$P_c v_c = m R T_c$~~   
 ~~$\bar{v} = 1845 \frac{ft^3/lb}{lb \cdot mol \cdot R}$~~

State 2:  
 $T_2 = 17^\circ C = 290.15 K$   
 $P_2 = 0.80 bar = 0.8 \cdot 10^5 Pa$

$R = 8.3145$

$(1.3 \cdot 10^5) V_1 = (2 \cdot 10^{-4}) (8.3145) (300.15)$   
 $V_1 = 9.59 \cdot 10^{-7} m^3$   
 ~~$V_1 = 9.59 \cdot 10^{-7} m^3$~~   
 ~~$= 9.59 \cdot 10^{-9} cm^3$~~

$(0.8 \cdot 10^5) V_2 = (2 \cdot 10^{-4}) (8.3145) (290.15)$   
 $V_2 = 1.507 \cdot 10^{-6} m^3$   
 $= 1.507 \cdot 10^{-8} cm^3$

$V_1 = 9.59 \cdot 10^{-7} m^3$   
 $V_2 = 1.507 \cdot 10^{-6} m^3$   
 $R = 8.3145 \frac{J}{mol \cdot K}$

$R = \frac{8.3145}{4.003} = 2.077 \frac{kJ}{kg \cdot mol \cdot K} \div 1000 = 0.002077 \frac{J}{g \cdot mol \cdot K}$

8.  $CO_2$  in closed, rigid tank

$Q = 2.75 kJ$

State 1:  
 $T_1 = 25^\circ C \rightarrow 298.15 K$   
 $P_1 = 2.75 bar = 2.75 \cdot 10^5 Pa = 2.75 \cdot 10^5 Pa$   
 $V_1 = 0.05 m^3$

State 2:  
 $T_2 = 257^\circ C \rightarrow 532.15 K$   
 $U_2 = 14,622 kJ$

$(2.75 \cdot 10^5) (0.05) = m (8.314) (298.15)$   
 $m = \frac{(2.75 \cdot 10^5) (0.05)}{(8.314) (298.15)} = 5.547 kg$

$U_1 = 6,885 kJ$

$U_2 = 14,622 \cdot 5.547 = 81,082.34$

80%  
 OKE P&E are negligible

$V_1 = 6,885 \cdot 5.547 = 38,141.095$   
 $Q + W = \Delta U$   
 $2.75 + W = 42,917.39$   
 $W = 42,914.64 kJ$

$\Delta U = 81,082.34 - 38,141.095$   
 $= 42,941.24 kJ$